

M. PHARM
(SEM I) THEORY EXAMINATION 2022-23
MODERN PHARMACEUTICS

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 2 = 20**

- (a) Define contour design?
- (b) Brief about the two method of drug excipient interaction with examples?
- (c) Explain about the URS and DQ in validation.
- (d) What are the different solubility enhancement technique
- (e) What is Chi square test.
- (f) Explain production and planning control
- (g) What are similarity factors – f₂ and f₁
- (h) What is the different effect of friction
- (i) Brief about the industrial and personal relationship
- (j) Write the Heckel equation and draw the Heckel plots for porosity of tablet compression

SECTION B

2. Attempt any two parts of the following:**2 x 10 = 20**

- (a) Discuss in detail about the objective and policies about the GMP. Also explain about the building layout, equipment's and their maintenance for solid dosage form.
- (b) Discuss about the scope and merits of validation. Give detail about the ICH & WHO guidelines for calibration and validation of equipment's.
- (c) What are the different optimization techniques in pharmaceutical formulation and processing. Discuss factorial design and its application in pharmaceuticals.

SECTION C

3. Attempt any five parts of the following:**7 x 5 = 35**

- (a) Discuss in detail about the physics of tablet compression. What are the different distribution forces involved in compression.
- (b) Brief about the Higuchi and Peppas plot along with its application in pharmaceuticals.
- (c) Discuss manufacturing and evaluation of emulsion.
- (d) Give a brief account on material and inventory control in pharma industry.
- (e) Discuss in detail about the different government regulation involved in validation of a dosage form.
- (f) What do you understand by response surface methodology. Give its application in optimization of dosage form.
- (g) What is the different pharmacokinetic parameter required for determination of bioavailability.